

Strict Positive Square Energy of Every Bicyclic Graph

AI-generated manuscript; no human author is claimed

30 July 2026

Abstract

Let G be a finite simple connected graph with n vertices and $n+1$ edges. We prove the strict inequality $s^+(G) > n$, where s^+ is the sum of the squares of the positive adjacency eigenvalues. The 2-core is either a bicyclic cactus (a figure-eight or handcuff/dumbbell) or a theta; arbitrary rooted trees may be attached at its vertices. The cactus class is covered by the strict theorem for every nonunicyclic cactus. For the non-cactus class we give the analytic attached-theta argument: tree messages, an open-right-half-plane Sachs expansion, and exact endpoint packets controlling zero, one, or two hostile odd cycles. The delicate negative Sachs coefficient of a 0 (mod 4) cycle is handled by explicit factorizations. The phase charge theorem gives, for every nonbipartite attached theta,

$$s^+(G) - n \geq 5 - 2\sqrt{5} > 0.$$

The cited finite verifier checks packet identities and hostile mutations, but the proof is uniform and does not use a finite path-length census.

1 Statement and structural reduction

All graphs are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, set

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad D(G) = s^+(G) - s^-(G).$$

Since $\text{tr } A(G)^2 = 2|E(G)|$, a bicyclic graph satisfies

$$s^+(G) + s^-(G) = 2n + 2, \quad s^+(G) = n + 1 + \frac{1}{2}D(G). \quad (1)$$

Theorem 1.1. *Every finite simple connected graph G with $|V(G)| = n$ and $|E(G)| = n+1$ satisfies*

$$\boxed{s^+(G) > n}.$$

Recall that the 2-core is obtained by repeatedly deleting vertices of degree at most one.

Proposition 1.2 (Bicyclic core dichotomy). *The 2-core H of a finite simple connected bicyclic graph is exactly one of the following:*

1. *a cactus with two cyclic blocks, either sharing one cut vertex (figure-eight or tight handcuff), or vertex-disjoint and joined by a path (dumbbell or loose handcuff);*
2. *a theta graph, consisting of three internally vertex-disjoint paths of positive lengths between two distinct terminals, with at most one path of length one.*

Every component outside H is a tree attached to exactly one vertex of H .

Proof. The core is connected, has minimum degree at least two, and retains $|E(H)| - |V(H)| = 1$. Thus

$$\sum_{v \in V(H)} (\deg_H(v) - 2) = 2.$$

Either one vertex has degree four and all others degree two, or two vertices have degree three and all others degree two. The first case gives two cycles meeting at the exceptional vertex. In the second case suppress maximal degree-two paths. If a suppressed edge is a bridge, the remaining half-edges form one cycle at each endpoint and the bridge expands to the joining path. Otherwise the suppressed multigraph has three parallel terminal-to-terminal edges, which expand to a theta. Simplicity forbids two direct terminal edges. Reversing leaf deletion attaches rooted trees, each at one core vertex. $\square$

The cactus alternative follows from the following strict theorem, whose manuscript includes the rank-uniform DNN frontier and closure of its residual families.

Theorem 1.3 (Nonunicyclic cactus theorem [2]). *Every finite simple connected cactus of cyclomatic rank at least two satisfies $s^+(G) > |V(G)|$.*

We prove the attached-theta alternative in enough detail that the main result does not treat the research note as a black box.

2 Tree messages and the right half-plane

For positive activities $a = (a_v)$ and a graph J , define

$$Z_J(a) = \sum_{M \text{ matching of } J} \prod_{v \text{ unmatched by } M} a_v. \quad (2)$$

Orient each off-core branch toward its core root. The leaf recurrence gives

$$Q_{u \rightarrow v} = t + \sum_{w \text{ child of } u} Q_{w \rightarrow u}^{-1} > 0, \quad a_v = t + \sum_{u \text{ off-core at } v} Q_{u \rightarrow v}^{-1} \geq t. \quad (3)$$

Extracting branch partitions gives one factor $K(t) > 0$. It remains common when a Sachs cycle deletes a core root: the detached branches contribute their full partitions. Thus no cycle-deletion term carries a hidden quotient.

Normalize

$$\Psi_G(t) = i^{-n} \det(itI - A(G)) = \prod_{j=1}^n (t + i\lambda_j), \quad t > 0. \quad (4)$$

Its continuous argument tending to zero at infinity is $\Theta_G(t) = \sum_j \arctan(\lambda_j/t)$, and the signed Coulson identity is

$$D(G) = -\frac{4}{\pi} \int_0^\infty t \Theta_G(t) dt. \quad (5)$$

This follows by integrating the identity for $\lambda|\lambda|$, using $\sum_j \lambda_j = 0$, and integrating by parts.

Let theta paths P_0, P_1, P_2 join terminals x, y , with $\ell_j = |E(P_j)|$. If their parities agree, the graph and all its tree attachments are bipartite, so spectral symmetry gives $s^+(G) = |E(G)| = n + 1$. Otherwise label the two odd and one even cycles

$$C_1 = P_0 \cup P_1, \quad C_2 = P_0 \cup P_2, \quad C_e = P_1 \cup P_2. \quad (6)$$

Put

$$d_j = Z_{P_j - \{x, y\}}(a), \quad \varepsilon = \begin{cases} +1, & |C_e| \equiv 2 \pmod{4}, \\ -1, & |C_e| \equiv 0 \pmod{4}. \end{cases}$$

Call an odd cycle hostile when its length is $1 \pmod{4}$, favorable when it is $3 \pmod{4}$, and set $\eta_j = +1$ or -1 accordingly. Sachs expansion gives

$$\frac{\Psi_G(t)}{K(t)} = W = R + 2i(\eta_1 d_2 + \eta_2 d_1), \quad R = Z_H(a) + 2\varepsilon d_0. \quad (7)$$

No two theta cycles are vertex-disjoint, so no double-cycle term occurs.

Lemma 2.1 (Open right half-plane). *For every $t > 0$, $R > 0$. Thus the continuous argument of W is its principal argument in $(-\pi/2, \pi/2)$.*

Proof. Partition matchings first on the even cycle. The omitted-path packet below gives $Z_H(a) = z_e d_0 + E_e$, where $z_e = Z_{C_e}(a|_{C_e})$ and $E_e \geq 0$. The even cycle has two perfect matchings and its empty matching has positive weight, so $z_e > 2$. Hence

$$R = (z_e + 2)d_0 + E_e > 0 \quad (\varepsilon = +1), \quad R = (z_e - 2)d_0 + E_e > 0 \quad (\varepsilon = -1).$$

□

3 The exact omitted-path packet

For a list of positive numbers define the continuant

$$K() = 1, \quad K(c_1) = c_1, \quad K(c_1, \dots, c_r) = c_r K(c_1, \dots, c_{r-1}) + K(c_1, \dots, c_{r-2}).$$

It obeys

$$K(c_1, \dots, c_r)K(c_2, \dots, c_{r-1}) - K(c_1, \dots, c_{r-1})K(c_2, \dots, c_r) = (-1)^r, \quad (8)$$

the determinant identity behind terminal path elimination.

Let C be one theta cycle and let its omitted path be $Q : x = v_0, v_1, \dots, v_r = y$, with $c_j = a_{v_j}$. Define

$$\begin{aligned} D_Q &= K(c_1, \dots, c_{r-1}), \\ L_Q &= \begin{cases} 0, & r = 1, \\ K(c_2, \dots, c_{r-1}), & r \geq 2, \end{cases} & R_Q &= \begin{cases} 0, & r = 1, \\ K(c_1, \dots, c_{r-2}), & r \geq 2, \end{cases} \\ M_Q &= \begin{cases} 1, & r = 1, \\ 0, & r = 2, \\ K(c_2, \dots, c_{r-2}), & r \geq 3. \end{cases} \end{aligned}$$

Empty continuants are one. If $z_C = Z_C(a|_C)$ and superscripts indicate deleted terminals, partitioning by the first and last edges of Q gives the exact four-state identity

$$Z_H(a) = z_C D_Q + z_C^x L_Q + z_C^y R_Q + z_C^{xy} M_Q. \quad (9)$$

This includes a direct omitted edge ($r = 1$) and a two-edge path ($r = 2$), the two cases in which informal empty-path conventions are most dangerous.

For a hostile cycle of length q , activity monotonicity gives

$$z_C \geq Z_{C_q}(t) =: Z_q(t), \quad 0 < \text{Arg}(z_C + 2i) \leq \text{Arg}(Z_q(t) + 2i). \quad (10)$$

The comparator is the isolated weighted cycle after tree elimination, not the bare attached theta; the latter comparison need not hold.

4 Channel lemmas

If neither odd cycle is hostile, (7) gives $\text{Im } W = -2(d_1 + d_2) < 0$, and hence $\Theta_G(t) < 0$. The one- and two-hostile channels require exact control of the negative coefficient created when $|C_e| \equiv 0 \pmod{4}$.

Lemma 4.1 (One-hostile channel). *Suppose C_1 is hostile, C_2 is favorable, and $q = |C_1|$. Then*

$$\Theta_G(t) < \text{Arg}(Z_q(t) + 2i) \quad (t > 0). \quad (11)$$

Proof. Set $B = d_2$, $A = d_1$, and $Z_q = Z_q(t)$, so $\text{Im } W = 2(B - A)$. If $|C_e| \equiv 2 \pmod{4}$, applying (9) to C_1 gives

$$\begin{aligned} R - Z_q(B - A) &= (z_{C_1} - Z_q)B + Z_qA + z_{C_1}^x L_{P_2} \\ &\quad + z_{C_1}^y R_{P_2} + z_{C_1}^{xy} M_{P_2} + 2d_0 > 0. \end{aligned} \quad (12)$$

If $B - A \leq 0$, the phase is nonpositive. Otherwise division by $RZ_q > 0$ and Lemma 2.1 give $\arctan(2(B - A)/R) < \arctan(2/Z_q)$.

Suppose instead that $|C_e| \equiv 0 \pmod{4}$. The last term in (12) becomes $-2d_0$, so termwise positivity would be invalid. Define

$$F(a) = Z_H(a) - 2d_0 - Z_q(B - A).$$

We show $F(a) \geq F(t) > 0$ on $a_v \geq t$. Let $I_j = V(P_j) - \{x, y\}$. Differentiating a matching partition deletes the corresponding vertex, and hence

$$\partial_v F = \begin{cases} Z_{H-v}, & v = x, y, \\ Z_{H-v} + Z_q Z_{I_1-v}, & v \in I_1, \\ Z_{H-v} - Z_q Z_{I_2-v}, & v \in I_2, \\ Z_{H-v} - 2Z_{I_0-v}, & v \in I_0. \end{cases}$$

For $v \in I_2$, unite a matching of C_1 and a matching of $I_2 - v$, using no boundary edge. This is a coefficient-preserving injection into matchings of $H - v$, so $Z_{H-v} \geq z_{C_1} Z_{I_2-v} \geq Z_q Z_{I_2-v}$. Similarly, using C_e and $I_0 - v$ gives $Z_{H-v} > 2Z_{I_0-v}$ because $Z_{C_e}(a) > 2$. Every derivative is nonnegative, proving $F(a) \geq F(t)$.

For strict bare positivity put $a = \ell_0$, $b = \ell_1$, $c = \ell_2$, and let $K_r = K(t, \dots, t)$ (r entries), with $K_{-1} = 0$, $K_0 = 1$. The residues force a even and b, c odd. A two-image, uncovered-vertex-preserving matching injection gives coefficientwise

$$(K_{a+b} + K_{a+b-2})K_{b-1} \geq 2K_{a-1}. \quad (13)$$

Indeed, for each matching of the interior of P_0 , complete through P_1 in the two parity choices and shift each retained P_0 edge by one in the second image. The images are disjoint because exactly the second uses the first P_0 edge, and both add b matched edges, preserving every power of t . Formula (9) now yields

$$F(t) = Z_q K_{b-1} - 2K_{a-1} + K_{a-1} K_{b-1} > 0 \quad (c = 1),$$

while for odd $c \geq 3$ it yields

$$F(t) = Z_q K_{b-1} - 2K_{a-1} + 2K_{a+b-1} K_{c-2} + K_{a-1} K_{b-1} K_{c-3} > 0.$$

The first difference is nonnegative by (13); a remaining term is positive. Therefore $R > Z_q(B - A)$ also in the $0 \pmod{4}$ channel, and the preceding sign split proves (11). $\square$

Lemma 4.2 (Two-hostile channel). *If both odd cycles are hostile, then*

$$\Theta_G(t) \leq \text{Arg}(Z_{|C_1|}(t) + 2i) + \text{Arg}(Z_{|C_2|}(t) + 2i). \quad (14)$$

The sum on the right is its continuous product argument, not a principal argument reset when the real part of the product changes sign.

Proof. Write $z_j = Z_{C_j}(a|_{C_j})$, $B_1 = d_2$, and $B_2 = d_1$. The packet identity with the other path omitted gives

$$R = z_j B_j + E_j, \quad E_j = z_j^x L_Q + z_j^y R_Q + z_j^{xy} M_Q + 2\varepsilon d_0. \quad (15)$$

For $\varepsilon = +1$, clearly $E_j \geq 0$. For $\varepsilon = -1$, the residues force P_0 odd and P_1, P_2 even. Each omitted path Q has even length at least two, so $L_Q, R_Q \geq 1$. Retaining a matching of $P_0 - \{x, y\}$ and adding the unique perfect matching of the remaining even path injects coefficient-preservingly into matchings of $C_j - x$ and $C_j - y$. Consequently the delicate expression factors exactly as

$$\begin{aligned} E_j &= (z_j^x - d_0)L_Q + d_0(L_Q - 1) \\ &\quad + (z_j^y - d_0)R_Q + d_0(R_Q - 1) + z_j^{xy} M_Q \geq 0. \end{aligned} \quad (16)$$

This remains valid when P_0 is one edge, when Q has length two, and when $M_Q = 0$; no empty-state convention has been suppressed.

Thus $2B_j/R \leq 2/z_j$. Put $\theta_j = \arctan(2/z_j)$ and $\Theta = \arctan(2(B_1 + B_2)/R) \in (0, \pi/2)$. If $\theta_1 + \theta_2 < \pi/2$, then

$$\tan(\theta_1 + \theta_2) = \frac{2/z_1 + 2/z_2}{1 - 4/(z_1 z_2)} \geq \frac{2(B_1 + B_2)}{R},$$

so $\Theta \leq \theta_1 + \theta_2$. If the sum is at least $\pi/2$, the result is immediate. Moreover

$$(z_1 + 2i)(z_2 + 2i) = (z_1 z_2 - 4) + 2i(z_1 + z_2)$$

has continuous argument $\theta_1 + \theta_2 \in (0, \pi)$, passing through $\pi/2$ rather than jumping branches. Apply (10) to both cycles to obtain (14). $\square$

5 The phase charge theorem

Theorem 5.1 (Attached-theta phase charge). *Let G be a finite simple connected graph whose 2-core is a theta, with arbitrary finite rooted trees attached at arbitrary core vertices. If G is nonbipartite, then*

$$D(G) \geq -4(\sqrt{5} - 2), \quad s^+(G) - |V(G)| \geq 5 - 2\sqrt{5} > 0. \quad (17)$$

If G is bipartite, then $s^+(G) = |V(G)| + 1$.

Proof. For a hostile odd cycle C_q , $q \equiv 1 \pmod{4}$, its spectrum and (5) give

$$D(C_q) = -2(\sec(\pi/q) - 1), \quad \int_0^\infty t \text{Arg}(Z_q(t) + 2i) dt = \frac{\pi}{2}(\sec(\pi/q) - 1). \quad (18)$$

For $q \geq 5$, this charge decreases and is at most $\delta := \sec(\pi/5) - 1 = \sqrt{5} - 2$. There are at most two hostile cycles. The zero-hostile phase is negative, Lemma 4.1 charges one hostile cycle to its bare cycle, and Lemma 4.2 charges two to their sum. Thus $D(G) \geq -4\delta$. By (1),

$$s^+(G) - n = 1 + \frac{1}{2}D(G) \geq 1 - 2\delta = 5 - 2\sqrt{5} > 0.$$

The last inequality follows from $25 > 20$. The bipartite assertion was proved before (6). $\square$

Proof of Theorem 1.1. By Proposition 1.2, a cactus core makes G a rank-two cactus, covered by Theorem 1.3; a theta core is covered by Theorem 5.1. Both alternatives include arbitrary tree attachments. $\square$

6 Reproducibility and scope

The full derivation underlying Sections 2–5 is the research note [3]; the proof above reproduces its uniform argument and key lemmas. The exact verifier [4] checks the endpoint states, low-length conventions, bare injection, $0 \pmod{4}$ factorization, phase ledger, and rejection of seven hostile mutations. From the repository root, run it normally and with optimization:

```
python3 positive-square-energy/experiments/\
    arbitrary_attached_theta_phase_verifier.py
python3 -O positive-square-energy/experiments/\
    arbitrary_attached_theta_phase_verifier.py
bash scripts/build-paper.sh all-bicyclic-graphs
```

The verifier is corroborative only: uniform completeness comes from the continuant recurrences and matching injections, not a finite path census.

Scoped nonclaim. The theorem concerns finite simple connected graphs with exactly $n + 1$ edges. It does not claim a strict bound for unicyclic or disconnected graphs, or an all-densities strengthening. Indeed $s^+(C_4) = |V(C_4)| = 4$.

AI disclosure

This manuscript and its proof-object assembly were generated by an AI coding and mathematical reasoning system from the cited repository materials. No human authorship, review, or verification is claimed. The result should be assessed from the displayed argument and reproducible dependencies. No external publication was performed as part of this work.

References

- [1] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [2] Companion manuscript, *Positive Square Energy of Every Nonunicyclic Cactus*, 2026; `all-nonunicyclic-cacti/paper.tex`.
- [3] *Arbitrary attached theta: continuant phase reduction*, research proof note, 2026; `positive-square-energy/bicyclic-theta/arbitrary-attached-theta-phase-note.md`.
- [4] *Supporting fail-closed checks for the attached-theta phase proof*, 2026; `positive-square-energy/experiments/arbitrary_attached_theta_phase_verifier.py`.